

Profile

Mass spectrometrists with five years' experience in native and denatured top-down proteomics. Recent experience with membrane protein characterization and capillary electrophoresis as an in-line separation tool. Extensive experience with R-based data analysis for both intact and fragmentation-based quantification.

Education

PhD Candidate in Chemistry, Northwestern University
Evanston, IL

June 2017 – Present

B.S. in Chemical Biology, University of California, Berkeley
Berkeley, CA

August 2014 – May 2017
GPA: 3.4

Relevant Research and Professional Experience

06/2017 – Present – Northwestern University Chemistry Department, Evanston, IL

PhD Candidate, Kelleher and Vafabakhsh Groups

Advisor: Professor Neil Kelleher and Reza Vafabakhsh

- Characterized phosphorylation states of G protein-coupled receptors (GPCR) using both middle-down and top-down platforms (see Publications)
 - Developed stable mammalian cell lines and protein purification protocols for GPCR work
- Performed relative quantification of histone isoforms using capillary electrophoresis top-down mass spectrometry
- Fractionated and characterized native protein complexes by size-exclusion chromatography into native top-down mass spectrometry
- Performed a large statistical comparison in R of fragmentation propensities in denatured versus native top-down methods to improve current scoring techniques used in ProSight Native (see Publications)
- Prepared and was awarded an F31 and T32 for GPCR work (see Awards)
- Mentored and trained several graduate and undergraduate students in laboratory skills, writing, and R programming
- Taught undergraduate lab sections, managed mass spectrometry lab component, held office hours and graded for organic chemistry

08/2014 – 05/2017 – University of California, Berkeley, Berkeley, CA

Undergraduate Researcher, Chemistry Department

Advisors: Professor Daniel Nomura

- Performed mammalian cell culture and bottom-up mass spectrometry and analysis
- Performed activity-based protein profiling to identify potential drug-candidates from heterologous cell lines

Relevant Technical Skills and Proficiencies

- **Mass Spectrometry:** native and denatured top-down mass spectrometry characterization of intact proteins, mass spectrometry method development including fragmentation optimization (higher collisional dissociation and electron-transfer dissociation), micro and nanoflow liquid chromatography, native and denatured capillary electrophoresis of protein samples

- **Mass spectrometry data analysis:** familiar with several mass spectrometry data platforms including ProSight, TD Validator, Xcalibur, Qualbrowser, MagTran, mMass, UniDec, ProteoFinder, and Oso; extensive experience with data transformations, analysis, and figure generation with R
- **Recombinant DNA methods:** Gibson assembly, restriction cloning, QuikChange site-directed mutagenesis, PCR, sequencing and interpreting sequencing results, DNA electrophoresis, plasmid DNA extraction (Mini and Maxi-Prep), bacterial transformation (heat shock and electroporation)
- **Protein Biochemistry:** immunoprecipitation (IP) and purification of soluble and membrane proteins, enzyme and signaling kinetics via luciferase-based assays, protein electrophoresis, protein immunostaining, conjugation of nanobody to sepharose for custom IPs
- **Confocal Microscopy:** live cell imaging using Zeiss instrumentation, cell surface and permeable labeling of protein targets for fluorescence microscopy, image work up and quantification using Fiji
- **Cell culture:** mammalian and bacterial cell culture, lentiviral transduction, bacterial protein expression, chemical transfection
- **General:** detail-oriented; proficient at presenting and providing written updates; proficient with R, Endnote, Microsoft Office Suite, Fiji, and PowerPoint; trained in ethics, rigor, and reproducibility; fluent in English
- **Interpersonal:** long and short-distance collaboration, teamwork, leadership, mentoring, and instruction

Publications

1. **Ives AN**, Dunn HA, Afsari HS, Dos Santos Seckler H, Foroutan MJ, Chavez E, Melani RD, Fellers RT, LeDuc RD, Thomas PM, Martemyanov KA, Kelleher NL, Vafabakhsh R. *J Am Chem Soc.* **2022**, *144* (50), 23104-23114
2. **Ives AN**, Su T, Durbin KR, Early BP, Dos Santos Seckler H, Fellers RT, LeDuc RD, Schachner LF, Patrie SM, Kelleher NL. Using 10,000 Fragment Ions to Inform Scoring in Native Top-down Proteomics. *J Am Soc Mass Spectrom.* **2020**, *31*(7), 1398-1409.
3. Kafader JO, Melani RD, Schachner LF, **Ives AN**, Patrie SM, Kelleher NL, Compton PD. Native vs Denatured: An in Depth Investigation of Charge State and Isotope Distributions. *J Am Soc Mass Spectrom.* **2020**, *31*(3), 574-581.
4. Schachner LF, **Ives AN**, McGee JP, Melani RD, Kafader JO, Compton PD, Patrie SM, Kelleher NL. Standard Proteoforms and Their Complexes for Native Mass Spectrometry. *J Am Soc Mass Spectrom.* **2019**, *30*(7), 1190-1198.
5. Cline EN, Das A, Bicca MA, Mohammad SN, Schachner LF, Kamel JM, DiNunno N, Weng A, Paschall JD, Bu RL, Khan FM, Rollins MG, **Ives AN**, Shekhawat G, Nunes-Tavares N, de Mello FG, Compton PD, Kelleher NL, Klein WL. A novel crosslinking protocol stabilizes amyloid β oligomers capable of inducing Alzheimer's-associated pathologies. *J Neurochem.* **2019**, *148*(6), 822-836.
6. Roberts AM, Miyamoto DK, Huffman TR, Bateman LA, **Ives AN**, Akopian D, Heslin MJ, Contreras CM, Rape M, Skibola CF, Nomura DK. Chemoproteomic Screening of Covalent Ligands Reveals UBA5 As a Novel Pancreatic Cancer Target. *ACS Chem Biol.* **2017**, *12*(4), 899-904.

Presentations

1. Oral presentation given at the American Society of Mass Spectrometry Annual Meeting. Minneapolis, Minnesota. June 8th, 2022. **Ives AN**; Dos Santos Seckler H, Kelleher NL, Vafabakhsh R. *Resolving isomeric phosphorylation states of GPCR C-termini using capillary electrophoresis (CE-MS/MS)*.
2. Oral presentation given at the American Society of Mass Spectrometry Annual Meeting. Virtual. June 3rd, 2020. **Ives AN**, Su T, Dos Santos Seckler H, Schachner LF, McGee J, Fellers RT, Patrie SM, Kelleher NL. *Trends from >15,000 Assigned Fragment Ions in Top-Down Mass Spectrometry*.
3. Oral presentation given at Clinical and Pharmaceutical Solutions through Analysis Annual Meeting. Langhorne, Pennsylvania. October 30th, 2019. **Ives AN**, Su T, Dos Santos Seckler H, Schachner LF, McGee J, Fellers RT, Patrie SM, Kelleher NL. *Trends from >15,000 Assigned Fragment Ions in Top-Down Mass Spectrometry*.
4. Oral presentation given at Chicago Mass Spectrometry Day. University of Chicago, Illinois. July 19th, 2019. **Ives AN**, Su T, Dos Santos Seckler H, Schachner LF, McGee J, Fellers RT, Patrie SM, Kelleher NL. *Trends from >15,000 Assigned Fragment Ions in Top-Down Mass Spectrometry*.
5. Poster presentation given at the US Human Proteome Organization Annual Meeting. Chicago, Illinois. March 7th, 2023. **Ives AN**, Dunn HA, Afsari HS, Dos Santos Seckler H, Foroutan MJ, Chavez E, Melani RD, Fellers RT, LeDuc RD, Thomas PM, Martemyanov KA, Kelleher NL, Vafabakhsh R. *Middle-Down Mass Spectrometry Reveals Activity-Modifying Phosphorylation Barcode in a Class C G Protein-Coupled Receptor*.
6. Poster presentation given at the American Society of Mass Spectrometry Annual Meeting. Atlanta, Georgia. June 4th, 2019. **Ives AN**, Su T, Dos Santos Seckler H, Schachner LF, McGee J, Fellers RT, Patrie SM, Kelleher NL. *Trends from >15,000 Assigned Fragment Ions in Top-Down Mass Spectrometry*.
7. Poster presentation given at the Keystone Symposium G Protein-Coupled Receptors: An Odyssey from Structure to Function. Salt Lake City, Utah. April, 2022. **Ives AN**, Dunn HA, Afsari HS, Dos Santos Seckler H, Foroutan MJ, Chavez E, Melani RD, Fellers RT, LeDuc RD, Thomas PM, Martemyanov KA, Kelleher NL, Vafabakhsh R. *Middle-Down Mass Spectrometry Reveals Activity-Modifying Phosphorylation Barcode in a Class C G Protein-Coupled Receptor*.

Awards and Outreach

Ruth L. Kirschstein Predoctoral Individual National Research Service Award (F31) (02/2022 – 01/2023)

- F31MH129114-01

Chemistry of Life Processes Fellowship (08/2018 – 08/2020)

- NIGMS Chemistry-Biology Interface Training Program supported by T32GM105538

Steven A. Hofstadler Graduate Student Award (10/2019)

- Presented at Clinical and Pharmaceutical Solutions through Analysis Annual Meeting for outstanding abstract and session presentation

Letters to a Pre-Scientist Participant (09/2018 – 09/2020)

- Served as a pen pal with middle school students. The focus is to build personal connections and help demystify science for students from high-poverty schools

Ashley N. Ives

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PhD Candidate, Northwestern University Chemistry Department

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Northwestern Summer Research Opportunity Program Mentor (6/2019, 6/2022)

- Trained and mentored undergraduates in laboratory work and scientific presentation for eight weeks. Developed a project for the student and provided technical training, as well as professional support and development

References

Available upon request